SAMRAT TIDKE

Senior Consultant | Data Engineer | MLOps | GenAI
samrat.tidke@gmail.com | +91-9923069054 | Pune, India

PROFESSIONAL SUMMARY

Innovative and results-oriented **Senior Consultant** with over **17 years of cross-functional experience** delivering **data engineering, MLOps, and GenAI solutions** across banking, manufacturing, healthcare, and cybersecurity domains. Expert in architecting and implementing **cloud-native, production-grade platforms** on **Google Cloud Platform (GCP)** and **Azure**, with deep hands-on exposure to **ETL/ELT design, ML lifecycle automation, RAG (Retrieval-Augmented Generation), large-scale vector/graph search, and microservices orchestration**.
Strong background in **multi-agent system design, observability, governance, and CI/CD automation**; known for bridging data science, platform, and business teams to build solutions that are technically robust, cost-optimized, and aligned with organizational goals.

CORE COMPETENCIES & TECHNICAL SKILLS

- Data Engineering & Analytics:
  Apache Spark 3.x, PySpark, Google Dataproc, Databricks, BigQuery, Hive, Apache Iceberg, Trino, Kafka (Confluent), Apache Airflow, Cloud Composer, Snowflake, Delta Lake, MinIO, Azure Synapse Analytics
- MLOps & Model Deployment:
  Vertex AI, MLflow, Seldon Core, TensorFlow, Keras, FastAPI, Kubernetes, Docker, Jenkins, GitLab CI/CD, Azure DevOps, Evidently AI, Cloud Build, Model Monitoring, Drift Detection, CI/CD Pipelines
- GenAI & LLMOps:
  Agentic AI, Multi-Agent Systems, LangChain, LangGraph, Pinecone, Neo4j, Weaviate, Vector Databases, Retrieval-Augmented Generation (RAG), Re-Ranking, LLM-as-Judge, Prompt Engineering, OpenAI API, Azure OpenAI, Vertex AI
- Programming & Cloud Platforms:
  Python, SQL, Bash, Linux Shell Scripting, REST APIs, GCP (BigQuery, Cloud Storage, Cloud Run, Pub/Sub, IAM, VPC-SC), Microsoft Azure (Data Factory, Synapse Analytics), Terraform, Infrastructure as Code (IaC)
- Machine Learning & Analytics Techniques:
  Supervised and Unsupervised Learning, Regression, Classification, Clustering, Time Series Forecasting, Topic Modelling, Artificial Neural Networks (ANN), Recurrent Neural Networks (RNN), LSTM, Predictive Analytics, Model Explainability, Bias Detection
- Practices, Governance & Tools:
  Agile, Scrum, DataOps, MLOps, Git (branching strategies), JIRA, Confluence, Documentation Standards, Cost Optimization, Observability, Compliance, Change Management

---

PROFESSIONAL EXPERIENCE

Atos India (Syntel Private Limited)– Senior Consultant | 23/12/2021 – Present

AI-Driven Incident Management | Agentic AI

- Built an MVP **AI-driven incident management platform** using **Agentic AI, multi-agent systems, and LLMs** to automate incident triage, RCA, remediation planning, approvals, and controlled execution across cloud and data platforms.
- Designed a **multi-agent RAG architecture** with **LangGraph, Lang Chain, Weaviate/FAISS, Pinecone, and Neo4j**, enabling semantic incident similarity, historical fix discovery, hybrid retrieval (vector + graph), and re-ranking.
- Implemented a **central orchestration layer (Main Agent)** coordinating **Platform Agent, Data Agent, and AirflowAgent** for intelligent classification, context enrichment, risk scoring, and remediation generation.
- Developed **re-ranking and script-matching pipelines** combining semantic similarity, metadata signals, graph history, and safety/blast-radius assessment to select **Terraform modules and Ansible playbooks**.
- Automated **governed remediation execution** via **GitHub Actions**, with pre-execution validation, **human-in-the-loop approvals**, rollback, and verification.
- Applied **LLM-as-Judge** for classification accuracy, RAG relevance, script safety, and remediation correctness, reducing false positives in production.
- Integrated **ServiceNow, Jira, GitHub Issues, Airflow APIs, and GCP services** using **Model Context Protocol (MCP)** for standardized agent-to-tool interactions.
- Deployed on **Kubernetes (Docker, GKE)** with **Langfuse observability**, autoscaling, health probes, audit trails, and measurable MTTR improvement.

Client: C1 Bank – Scalable Data Pipelines & Lakehouse

- Led the **architecture and build of real-time, batch, and scheduled pipelines** with **Apache Airflow** and **Spark**, ingesting from **Kafka (Confluent)** and landing data in **MinIO**.
- Implemented **streaming ETL** with **Spark Structured Streaming, Trino, and Iceberg**, supporting federated queries across structured/unstructured data.
- Modelled enterprise datasets in **Snowflake (Data Vault 2.0)** and promoted **Data Mesh** to deliver business-aligned, self-serve data products.
- Automated **data quality & anomaly checks** using **Great Expectations**; enforced versioning and governance via **Git/GitLab**.
- Built **CI/CD pipelines** (GitLab runners) for Spark streaming jobs, delivering faster releases and improving developer productivity.
- Containerized jobs and orchestrated deployment schedules, integrating results into operational dashboards and analytics platforms.

Client: Johnson & Johnson – ML & MLOps Pipelines

- **Developed and maintained complete ML workflows** for ingestion, preprocessing, feature engineering, training, evaluation, and packaging using MLflow, **Vertex AI Pipelines**, **Cloud Composer (Airflow)**, and **Dataproc/Spark**.
- Automated **training, deployment, and online serving** on **Vertex AI Endpoints**, introducing **event-driven retraining, canary deployments, and rollback protocols** to protect production stability.
- Established **CI/CD automation** via **Cloud Build and GitHub Actions**, embedding testing, linting, artifact management, and environment promotion rules.
- Designed an **observability stack**—Grafana dashboards, Evidently AI for drift, and Seldon Core for model metrics—allowing quick identification of performance degradation or data drift.
- Strengthened **governance and coding discipline**: standardized branching, implemented model registry policies, documented pipeline lifecycles, and led retrospectives after incidents.
- Partnered closely with data scientists, product owners, and platform engineers to convert PoCs into **robust, SLA-backed ML services**, enabling new predictive capabilities.

Productiva IT Solutions – GCP Senior Data Engineer | (08/04/2021 – 22/10/2021)

- **Engineered ETL pipelines** with **Dataflow (Beam)** and **Dataproc (Spark)** to ingest, normalize, and enrich data. Optimized schema handling and replay logic for late or malformed data.
- Automated **batch and streaming dataflows** using **Cloud Composer** and **Jenkins CI/CD**, improving release agility and cutting manual steps.
- Designed a **secure data lake and analytics platform** using **Google Cloud Storage and BigQuery**, applying tiered storage, lifecycle rules, and table partitioning for performant and compliant threat analytics.
- Collaborated with security operations and analytics teams to tune **BigQuery partitioning and clustering**, reducing forensic query latency and storage costs.
- Authored knowledge assets—architecture docs, runbooks, onboarding guides—empowering teams to add new log sources or troubleshoot pipelines independently.

IAV India Pvt Ltd – Data Engineer | (*01/07/2014 –31/12/2020)

- **Developed comprehensive ingestion and preprocessing pipelines** for large-scale automotive sensor and test-bench data, implementing validation, cleaning, and feature extraction routines to support ML model training.
- Executed **statistical analyses, predictive modelling, and trend studies** to detect anomalies, forecast equipment wear, and recommend proactive maintenance strategies.
- Built **real-time analytics services** streaming processed data into **Hive** and **Cassandra**, enabling engineers to query live metrics and visualize results with minimal latency.
- Designed **bespoke dashboards** (Python / matplotlib / Plotly, SQL, Excel) to present KPIs, durability results, and system health for R&D stakeholders.
- Partnered with mechanical engineers, data scientists, and QA specialists to codify domain expertise into repeatable pipelines, improving documentation and lab-wide data governance.

Mechanical Engineer | (12/02/2009 – *30/06/2014)

- Designed automotive components and sub-systems using CAD and engineering simulation tools; applied statistical methods to optimize material strength and reduce cost.
- Collaborated with cross-functional teams (design, testing, production) to validate prototypes and achieve performance standards.

Magna Steyr India – Engineer | (21/01/2008 – 10/02/2009)

- Project co-ordination & documentation for offshore automotive programs.

Access CADD   - Engineer |(04/06/2007 – 14/01/2008)

- Mahindra Vendor Development Engineer (process quality & analytics improvements).

Seam Arc Welding Products – Design Engineer | (20/05/2005 – 25/05/2007)

- Manufacturing optimization, reporting automation.

EDUCATION

- **B.E. Mechanical Engineering**, K.K. Wagh College of Engineering, Nashik (1998-2004)
- HSC – K.T.H.M. College, Nashik (1996-1998)
- SSC – RJEMS High School, Nashik (1996))